

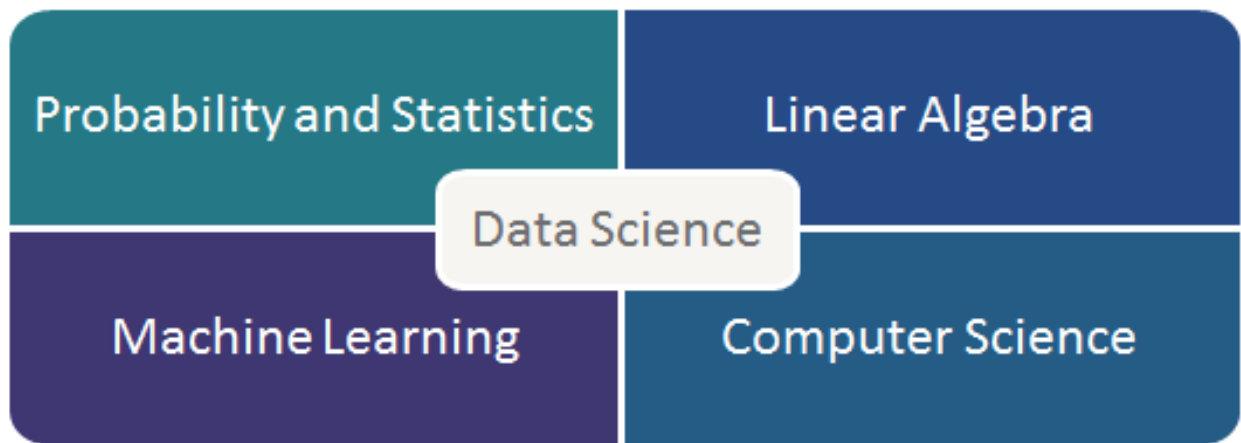
Introduction to Data Science

Data Science

Data Science is an interdisciplinary field about processes and systems to extract knowledge or insights from data in various forms, either structured or unstructured

Data Science is the empirical synthesis of actionable knowledge from raw data through the complete data lifecycle process

Components of Data Science



1. Probability and Statistics

Probability

Probability is a mathematical subject which enables us in determining or predicting how likely it is that an event will happen.

The probability of occurrence of an event is assigned a value between:

$$0 \leq P(E) \leq 1$$

Where:

- $(P(E) = 1)$ means the event will occur with complete certainty.

- $(P(E) = 0)$ means the event is impossible.
- Higher probability values indicate a greater likelihood of occurrence.

Statistics

Statistics is a branch of mathematics that deals primarily with data.

It involves methods for:

- Collecting data
- Organizing data
- Classifying data
- Presenting data
- Analyzing and interpreting data

The analysis of refined data helps in drawing meaningful conclusions and insights.

2. Linear Algebra

Linear Algebra is a mathematical subject that deals with the theory of systems of linear equations, matrices, vector spaces and linear transformations.

Linear Algebra is critically used in almost all peripheries of science, practically solving most of the problems using linear models.

Most of the complex science problems are converted into problems of vectors and matrices and then solved using linear models.

In the world of data (especially, big data), linear algebra can be very handy to process huge chunks of data to accomplish many practical transformations.

Iterative Method vs Linear Algebra Method

Iterative Method	Linear Algebra Method
Uses loops for computations	Uses optimized linear algebra functions
More time-consuming	Much faster and efficient
Example elapsed time: ~15.04 seconds	Example elapsed time: ~0.038 seconds

#Solving using Iterative Method

```
No_Cols <- 10000 # No. of Columns  
No_Rows <- 1000 # No. of Rows
```

#Computation 1

```
sum <- 0.0  
for(i in 1:No_Rows){  
  for(j in 1:No_Cols){  
    sum1 <- sum1 +  
      as.numeric(mat1[i, j])^2  
  }  
}  
frobenius_norm <- sqrt(sum1)
```

#Elapsed Time: ~ 15.04 seconds

#Solving using Linear Algebra

```
No_Cols <- 10000 # No. of Columns  
No_Rows <- 1000 # No. of Rows
```

#Computation 2

```
frobenius_norm <-  
  norm(as.numeric(mat1),type="F")
```

**#Note: In the above code norm()
is a function in R which uses
Linear algebra internally.**

#Elapsed Time: ~ 0.038 seconds

Linear algebra technique is much more efficient than the iterative method.

3. Computer Science

Computer Science provides us with the necessary programming languages, database management systems, statistical analysis and machine learning tools.

4. Machine Learning

A field of study that gives computers the ability to learn without being explicitly programmed.

Types of Machine Learning

Supervised Learning Used in applications when labelled historical data predicts likely future events.	Unsupervised Learning Used when historical data is not labelled and is used to discover unknown patterns in the data.
Semi-supervised Learning Used in same applications as supervised learning. Uses both labelled and unlabeled data for training.	Reinforcement Learning Works by discovering which action yields maximum reward through trial and error. Often used in robotics, gaming and navigation.

4.1. Supervised Machine Learning

Learning Phase

A machine is taught to identify various fruits by building a model with the help of images.



Apple



Apricot



Banana



Pear



Grapes



Orange

Testing Phase

A new set of images is given to this model as test data so that it can classify different fruits.



Evaluation Phase

We should always evaluate a model to know whether it will do a good job of predicting the target, given new data points.

One way to do so is to compute the ratio of correctly classified test data points to the total number of test data points available, thus determining the accuracy of the model.

In our example, out of eight test images, machine was able to classify 6 images correctly and 2 images incorrectly.

Hence, accuracy of this supervised machine learning model is:

$$\frac{6}{8} = 75\%$$

Types

There can be two types of supervised machine learning techniques as shown below:

Classification

Used to predict discrete results.

For example, assume that a company wants to predict the budget period of a new project that they have acquired as:

- Short-term
- Long-term

based on various input attributes about the project such as:

- Number of resources required
- Software requirements
- Hardware requirements

We will need to use the classification machine learning technique here.

Regression

Used to predict continuous numeric results.

For example, if we are trying to predict the approximate budget requirement of a new project that the company has acquired in actual quantifiable figures, based on various input attributes about the project such as:

- Number of resources required
- Software requirement
- Hardware requirement

then we use the regression technique.

4.2.Unsupervised Machine Learning

There is a basket filled with some fresh fruits.

The machine's task is to group similar colored fruits together.

Clustering

Machine identifies four clusters of fruits based on their color, as shown below.



- In the first step, we fix some variables or parameters based on which the machine will arrange the given data (in our example, we have taken "color" as the parameter).
 - In the second step, the machine groups similar data points together. In our example, fruits with the same color are grouped together.
-

Semi-Supervised Machine Learning

In real-time, it may so happen that the unlabeled data points exceed the number of labeled data points in a data set.

In order to fit a model to such data we use the semi-supervised machine learning technique, wherein we perform the following steps:

Step 1

Train the model with labeled data points only.

Step 2

Use the above model to predict the labels of the unlabeled data points.

Step 3

Combine the existing labeled data points with the newly labeled data points and use it to retrain the model.

Step 4

Repeat the 2nd and 3rd steps until it converges.

Applications of semi-supervised learning are:

- Text processing
 - Video-indexing
 - Bioinformatics
 - Web page classification
 - News classification
-

Reinforcement Learning

Reinforcement machine learning algorithm is a reward based and immediate feedback technique.

Here, the machine's goal is to maximize the numerical reward at each and every step.

In the process of learning, the machine is not provided any supervision as opposed to the previous ML algorithms we discussed till now.

Instead, the machine is expected to figure out the optimum actions which will reap the maximum reward at each step, all on their own, without any interference.

The actions that the machine takes at each step might not only affect the immediate reward but may also affect all the subsequent rewards.

The ultimate aim is to reach the max possible reward in the least amount of steps possible.

Thus, trial and error search methodology and immediate feedback in the form of a numerical reward are the two main characteristics of reinforcement learning.

Examples

- A machine learning to play chess decides whether a move is right by planning the possible moves, anticipating the corresponding counter moves and finally choosing one based on reward based appeal for a particular position or set of moves.
- A trash collecting bot whose charge is about to reach critical levels needs to decide whether to clean one more room before reaching out for the charging station or to immediately rush to the nearest charging station.

The decision taken by the bot depends on the ease with which it can reach the charging station, based on its prior knowledge.

Machine Learning Algorithms

Supervised Learning

- Linear regression
- Logistic regression
- Naive Bayes
- K-Nearest Neighbors
- Decision trees
- Random Forests
- Support Vector Machines

Unsupervised Learning

- k-means Clustering
- Hierarchical Clustering
- Mixture Models

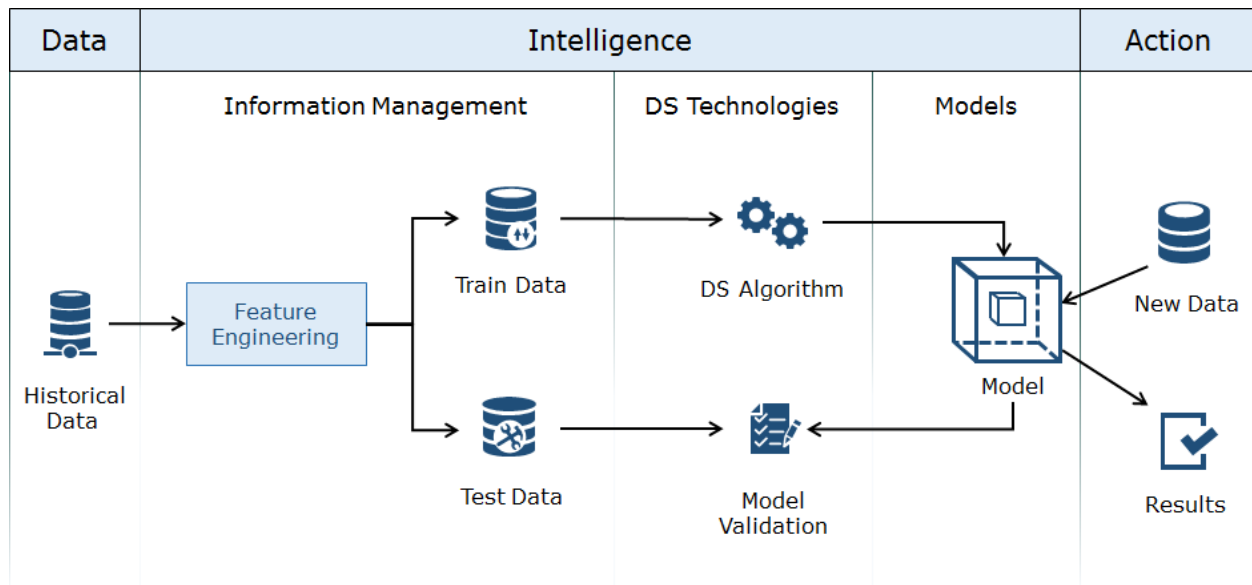
Semi-supervised Learning

- Graph-based methods
- Generative models
- Low-density separation
- Heuristic approaches

Reinforcement Learning

- Markov decision processes
- Monte Carlo Methods
- Temporal-Difference Learning

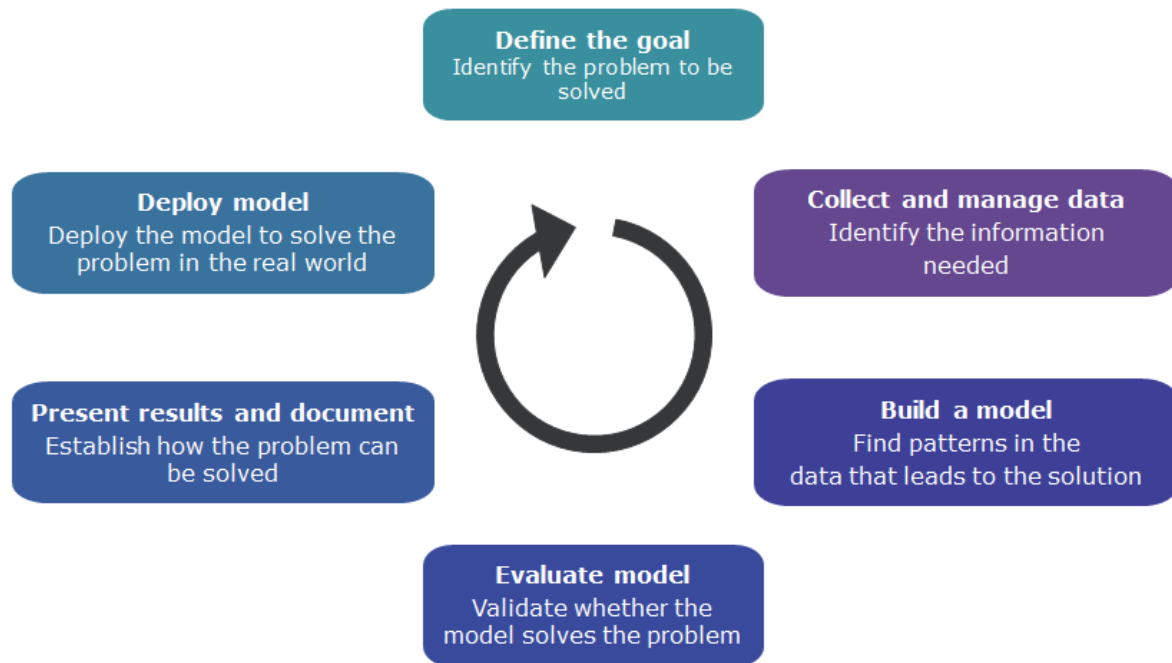
Data Science Process



Data Science Architecture

Data	Intelligence			Action
 Data Sources  Apps  Sensors and Devices	Information Management - Databases - Spark - Hadoop	DS Technologies - Exploratory Data Analysis - Statistical Inference - Association Analysis - Regression Analysis - Time Series Analysis - Machine learning	Models 	 People  Web  Apps  Bots  Mobile  Automated Systems
- Historic Business data that can be used to gain business insights. - It can lead to gain in efficiency	- Construct and select features for the problem - Cleaning the data - Distributed/parallel computing environment for handling massive amounts of data	Contains components that create a model based on the patterns in the training data.	- Is the pattern/knowledge extracted from the data - Given new data, it produces the desired result (Classification, regression etc)	Layer that uses the model to gain business insights.

Data Science Project Life Cycle



Data Science Implementation - Business Use Case

Country Bank of India wants to cut down on their losses due to bad loans.

It approaches a data analytics firm to help them reduce these losses by X%.

Step 1: Define the Goal

The first and foremost step in any project is to define a clear goal.

Hence, at this point, it is important to learn every minute detail about the project such as:

- Why is the project being started?
- What is missing currently and what exactly is required?
- What are they currently doing to fix the problem, and why isn't it working?

- What all resources are needed?
- What kind of data is available?
- Is domain expertise available within the team?
- What are the computational resources available/required?
- How does the business organization plan to deploy the derived results?
- What kind of problems need to be addressed for successful deployment?

Bad Loan Use-Case: Define the Goal

The goal is to lessen the bank's losses caused by bad loans.

To do this the firm intends to create a tool to help the bank's loan officers improve their accuracy in identifying bad loan applicants, thereby lowering the number of bad loans being authorized.

For this purpose, the goal defined should be to the point and unambiguous.

For example:

"We want to reduce the rate of loan charge-offs by at least 10%, using a model which predicts whether loan applicants are likely to default"

is preferred over:

"We want to get better at finding bad loans"

Step 2: Collect and Manage Data

Now that the goal has been set, the next step is to find, explore and clean the data necessary for analysis.

This stage takes up a lot of time but helps in finding answers to many important questions, such as:

- What all data is available?
- Will it help in solving the problem?
- Is the data enough to carry out analysis?
- Is the quality of data up to the mark?

Bad Loan Use-Case: Collect and Manage Data

- Collect the data about each loan application with relevant attributes such as:
 - Status of loan
 - Duration
 - Credit history of the applicant
 - Present employment status
 - Residing at an address since
 - Number of dependents
 - Number of active loans under the applicant's name

- Collect the data across a reasonable span of time such as one year or one decade.
- Conduct initial exploration (using data visualization and summary statistics) and clean the data.

While refining the data, it may so happen that the data identified earlier turns out to be not adequate to perform the analysis.

There might also be a situation wherein we encounter various new problematic areas within the data, which we disregarded as not being a problem at all, previously.

For example, if the data set we took contained most of the defaulters or just a few defaulters, our analysis may result in a biased conclusion.

Step 3: Build a Model

Once the data is ready, the next step is to find meaningful insights from the data.

Depending on the nature of the business problem we are dealing with we can make use of any of the following data modelling techniques to gather such insights.

- **Classification:** Determining which among the given categories a data point falls under
- **Scoring:** Predicting or estimating a quantifiable value
- **Ranking:** Ordering the data points depending on the priorities involved
- **Clustering:** Grouping similar items based on certain parameters
- **Finding Relations:** Finding associations between various features of the data
- **Characterization:** Creating plots, graphs and various reports for understanding the data better

Bad Loan Use-Case: Build a Model

In the bank scenario, the problem we are dealing with is classification.

We wish to classify bank customers who apply for loans as:

- Probable defaulters
- Non-defaulters

Hence, we need to train our model in such a way that it covers the entire range of the available data, thus enabling it to learn about most of the probable loan defaulter cases.

Given the preceding requirement, we decide on a suitable approach to build the model.

We can choose from:

- Logistic regression
- Naive Bayes
- K-nearest neighbours
- Decision trees

among other available classification techniques.

We also need to be aware of why a model is taking a particular decision and how confident it is in its prediction.

Ultimately, our model should be able to answer the question:

"How likely is an applicant to be a defaulter?"

Step 4: Model Evaluation

Now that we have built our model, we need to determine whether it meets our goals by asking the following questions:

- Is the model accurate enough for our needs?
- Does the model meet the expectations?
- Is it better than the methodology being currently used?

If for either of the above questions, the answer is **NO**, we need to revisit the previous steps.

Bad Loan Use-Case: Model Evaluation

- Check whether the evaluation parameters from the suggested model apply in our scenario.
 - Proceed to calculate the model evaluation parameters (such as accuracy and precision) based on the predefined rules and observe how many predicted values match the actual values.
-

Step 5: Present Results and Document

At this stage we have achieved a desirable model.

A model that meets all the requirements and goals we set for ourselves at the beginning of the project.

The next step is to showcase the project to various audience as follows:

- Present the details of the model to all the collaborators, clients and sponsors.
- Provide everyone in charge of usage and maintenance of the model, once deployed, with documentation that covers all aspects of the working of the model.

We must keep in mind that each group of people involved in the project require different kind of treatment, when it comes to presentations and providing documentation.

Hence, specific data visualization techniques must be used for each of them.

What might work for one audience, might not work for the other.

Bad Loan Use-Case: Present Results and Document

In the bad loans scenario, we need to showcase the efficacy of our model to the bank officials.

For example, if our model is able to identify a set of bad loans that amount to 36% of the total money lost to defaulters, we need to emphasize the same in the executive summary.

In addition to this, any interesting insights uncovered such as:

- New auto loans are more riskier than used auto loans
- Less losses are tied to home loans

must also be presented.

Step 6: Deploy Model

The last and final step is to deploy the model.

Usually from this point ahead the data scientist is no longer associated with the operations of the model.

But before they are off the job they must make sure that the following are in place:

- The model has been tested thoroughly and generalizes well.
- The model should be able to adjust well to unforeseen environmental changes.
- The model has been deployed in a pilot program and any problems that cropped up in the last moment were taken care of by updating the model accordingly.

Bad Loan Use-Case: Deploy Model

There may arise a situation wherein experienced loan officers might veto the decision taken by the model that we created as it opposes their instincts.

Hence, we need to be always on the look out for which is correct, our model or their intuition?

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